

How to Make Health Decisions Without Data

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LONGEVITY
HEALTH CLINIC

Randomized Clinical Trials

1. Does it work on average?
2. Is it safe enough to try?
3. Is it working or will it work in this person?

No large randomized trial → no guideline recommendation → insurance doesn't cover it → most physicians don't prescribe

GLP-1s

Diabetes ~\$250M

→ Obesity ~\$200M

→ Cardiovascular event reduction ~\$350M

→ Sleep apnea ~\$50M

→ MASH ~\$100M

\$950 MILLION in randomized clinical trials

GLP1s and statins

The 2026 ACC/AHA dyslipidemia guideline still describes lifestyle management as foundational and gives healthy diet a Class 1A recommendation for cardiovascular-risk reduction

GLP-1s and alcohol

Excessive alcohol consumption contributes to roughly 178,000 U.S. deaths annually, according to CDC estimates for 2020–2021—a 29% increase from 2016–2017

2025 randomized JAMA Psychiatry study of 48 people with AUD: Semaglutide at low doses reduced alcohol consumption, drinks per drinking day, craving and heavy drinking over time

GLP-1s and safety

Risk of Nonarteritic Anterior Ischemic Optic Neuropathy in Patients Prescribed Semaglutide

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Key Points

Question Are prescriptions for semaglutide associated with an increased risk of nonarteritic anterior ischemic optic neuropathy (NAION) in patients with type 2 diabetes or patients who are overweight or obese?

Findings This matched cohort study of 16 827 patients revealed higher risk of NAION in patients prescribed semaglutide compared with patients prescribed non-glucagon-like peptide receptor agonist medications for diabetes or obesity.

Meaning The findings suggest a potential risk of NAION associated with prescriptions for semaglutide, but future study is required to assess causality.

Testosterone Replacement in Men

- loss of libido
- declining muscle mass
- fatigue
- impaired recovery
- loss of motivation
- poor sleep

298

302

TRT guidelines

The Testosterone Trials enrolled 790 symptomatic men 65 or older whose testosterone was below 275 ng/dL. Testosterone improved sexual activity, sexual desire and erectile function, with different effects across the other measured domains

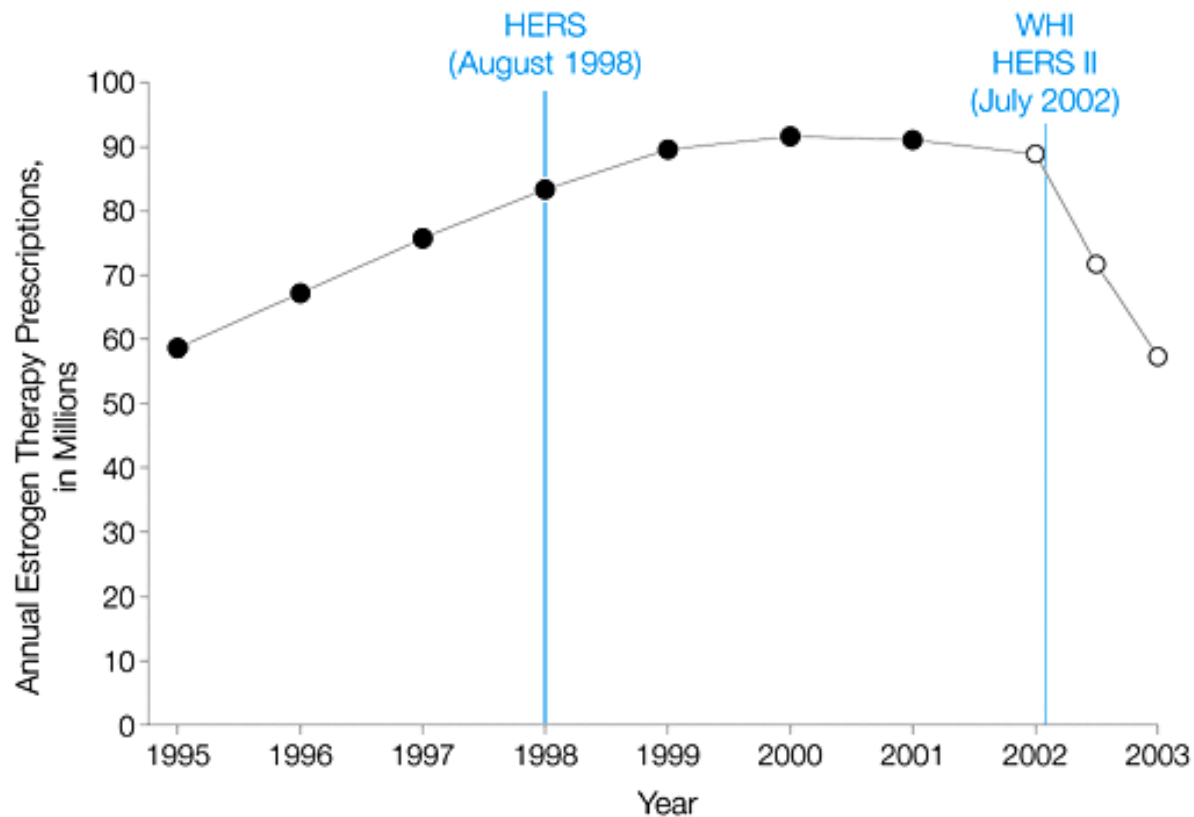
TRAVERSE then enrolled 5,246 men with symptoms and two fasting testosterone levels below 300 ng/dL to address cardiovascular safety

We define the eligible population as testosterone below 300.

Then we say:

“There are inadequate clinical-trial data supporting treatment above 300.”

Menopause and HRT



Cost of clinical trials

Trials requiring actual clinical outcomes rather than surrogate markers averaged about \$65 million

GLP-1s: Ozempic, Wegovy, Rybelsus, Victoza, Saxenda have 2025 worldwide sales of \$35 billion

Medications like testosterone and BPC-157 cannot be patented

How Do We Make Decisions Without RCTs?

What exactly am I trying to change - what are the outcomes I want to achieve?

“My elbow pain is 7/10 when lifting”

“I drink three alcoholic drinks every evening”

“I wake up four times per night”

“My 10 year heart attack and stroke risk is 20%”

“My libido has fallen from 8/10 to 2/10”

***Wearable data, bloodwork

Mechanism

Mechanism

- cell studies
- animal studies
- observational/uncontrolled human evidence
- small randomized trials
- large randomized trials
- long-term outcomes

Safety Signals

NAD⁺ and cancer

- NAD⁺ in a petri dish with cancer cells causes them to grow

BPC-157 and cancer

- Preclinical pro-angiogenic effects involving VEGFR2 signaling.

Using AI

Do GLP-1s cause medullary thyroid cancer?

Review the case reports of medullary thyroid cancer occurring in patients taking GLP-1s and tell me if GLP-1s cause MTC?

Do GLP-1s cause acute pancreatitis?

Review meta-analyses of GLP-1s and tell me if GLP-1s cause acute pancreatitis

Bayesian Reasoning for BPC-157

What do I think the probability is that this works before I try it?

-Right shoulder pain 8/10 limiting bench press $<X$ and left quad tendon injury/pain 8/10 limiting squats $<Y$

-Prior MRI, months avoiding exercise, prior steroid injection, surgery offered: none have provided durable results

Probability of Success: 20%

Intervention

BPC-157 stacked with tesamorelin/ipamorelin in the evening M-F, off Sat-Sun

- Within 1 week, pain reduced to 2/10 at both sites and new PRs for bench press, squats and leg presses

- Results durable over the second week, no new injuries

Washout

Discontinue the BPC/tesa/lpa stack for 1-2 weeks

-Pain increased to 6/10 within 1 week of stopping

Probability of Success: 90%

N-of-1 trial

BASELINE → TREATMENT → WITHDRAWAL → RECHALLENGE

1. Define outcomes and timeframe
2. Test one intervention at a time
3. Washout period
4. Reassess

The question changes from “does it work for most people” to “does it work for me”

Questions

Dr. Jason Itri is the founder of Longevity Health Clinic. With more than a decade of experience in providing holistic and empathetic care, Dr. Itri has established himself as a leader in the field of health and wellness.

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